

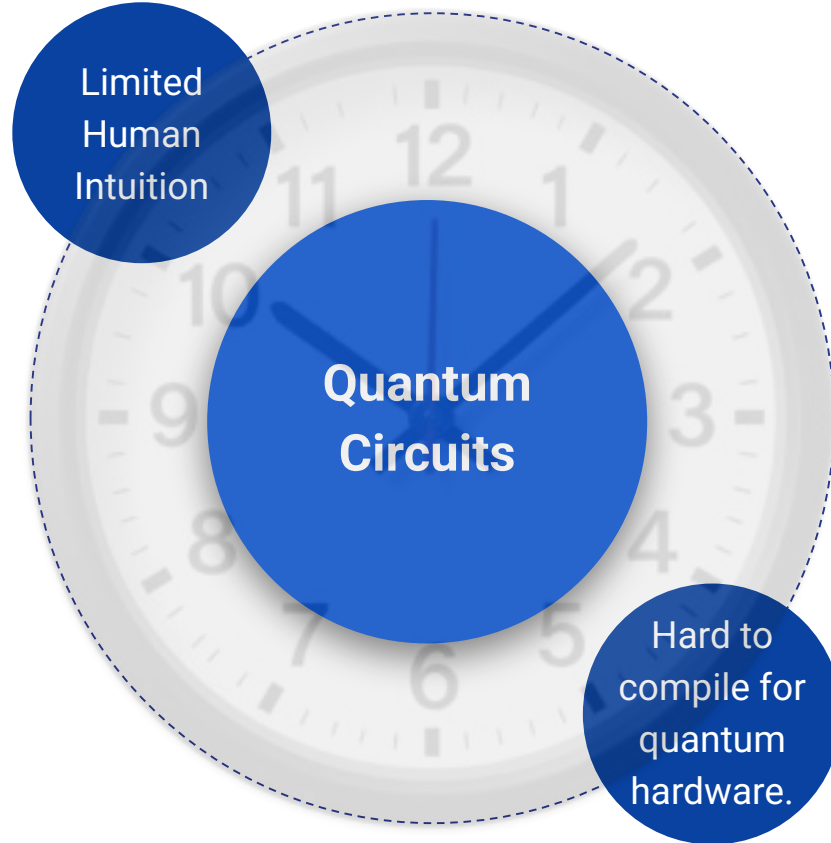
Case 1: MQS Xenakis

Team 8

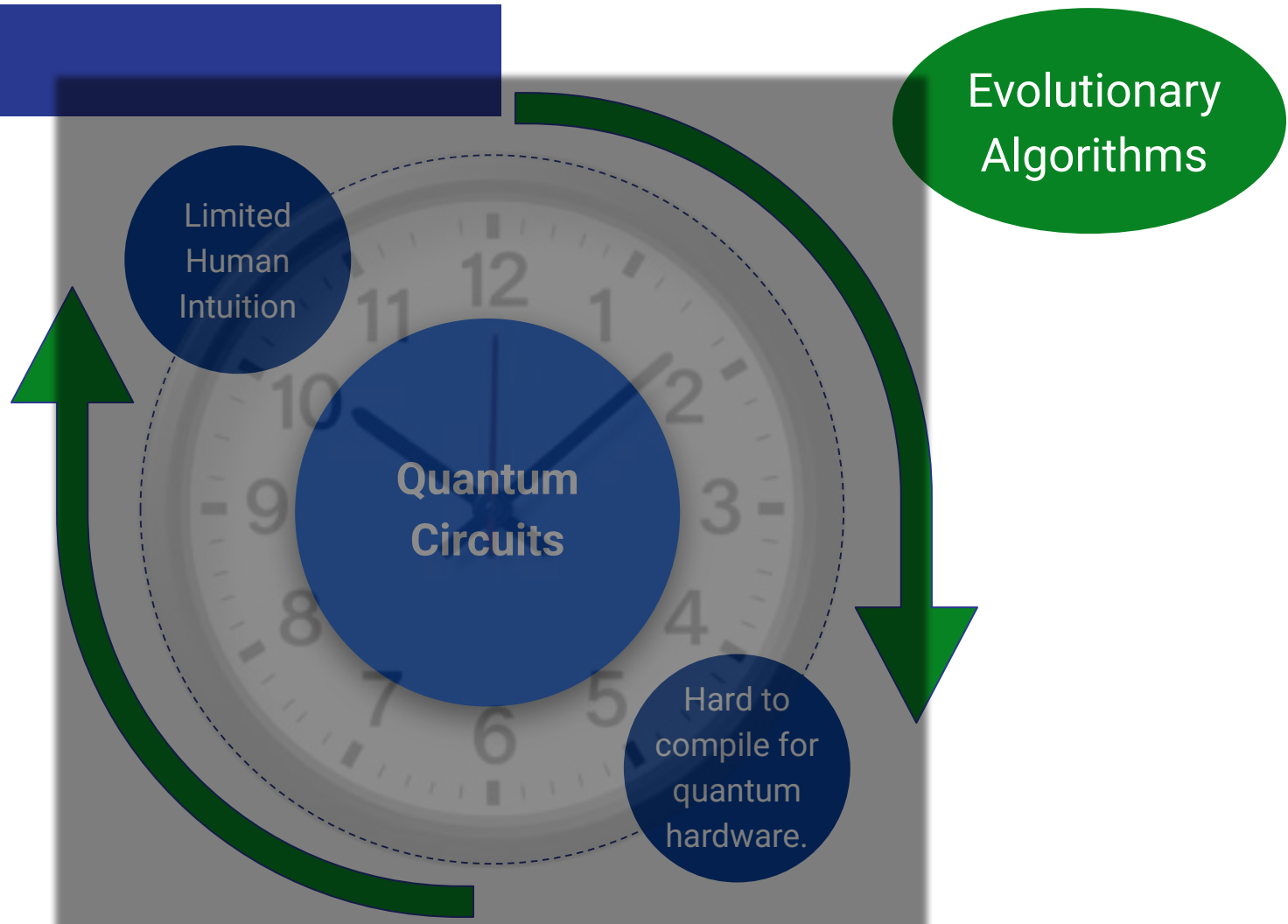
B. E. Castiblanco (Aarhus University)
J. C. Henao Montoya (Université de Strasbourg)
E. Muñoz (University of Copenhagen)
A. L. Cício (University of Copenhagen)



Problem



Solution

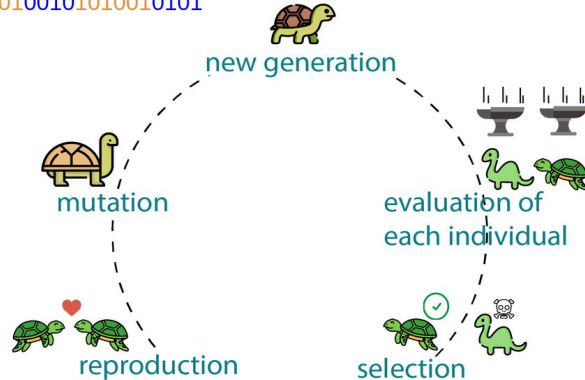
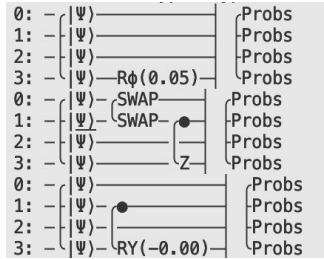


Solution

Xenakis

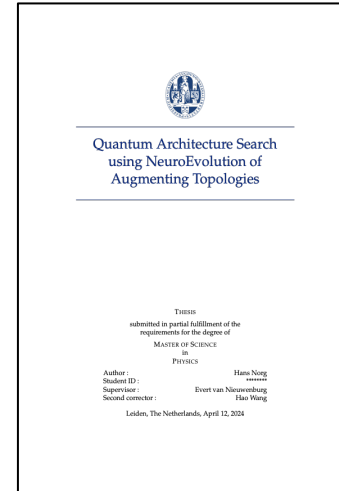
Genetic Algorithm

01010001010101010101 ... 10010100101010010101

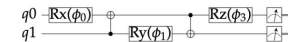
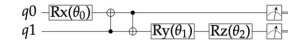


QASNEAT

QNEAT



Parent 1	1 Rx θ_0 0	2 CNOT - 0, 1	3 CNOT - 1, 0	4 Ry θ_1 1	8 Rz θ_2 1	
Parent 2	1 Rx ϕ_0 0	2 CNOT - 0, 1	3 CNOT - 1, 0	4 Ry ϕ_1 1	7 CNOT - 1, 0	10 Rz ϕ_2 0
Child	1 Rx θ_0 or ϕ_0 0	2 CNOT - 0, 1	3 CNOT - 1, 0	4 Ry θ_1 or ϕ_1 1	8 Rz θ_2 1	10 Rz ϕ_2 0



Solution

Xenakis

Genetic Algorithm

QASNEAT

QNEAT

Circuit for VQE

H H

The diagram illustrates a workflow for solving a problem using a Variational Quantum Eigensolver (VQE). It features two parallel paths that converge. The left path is labeled 'Xenakis' and 'Genetic Algorithm'. The right path is labeled 'QASNEAT' and 'QNEAT'. Both paths lead to a central point, from which a single path descends to a box labeled 'Circuit for VQE'. To the right of this box are two circular icons, each containing the letter 'H', representing Hadamard gates in a quantum circuit.

Results

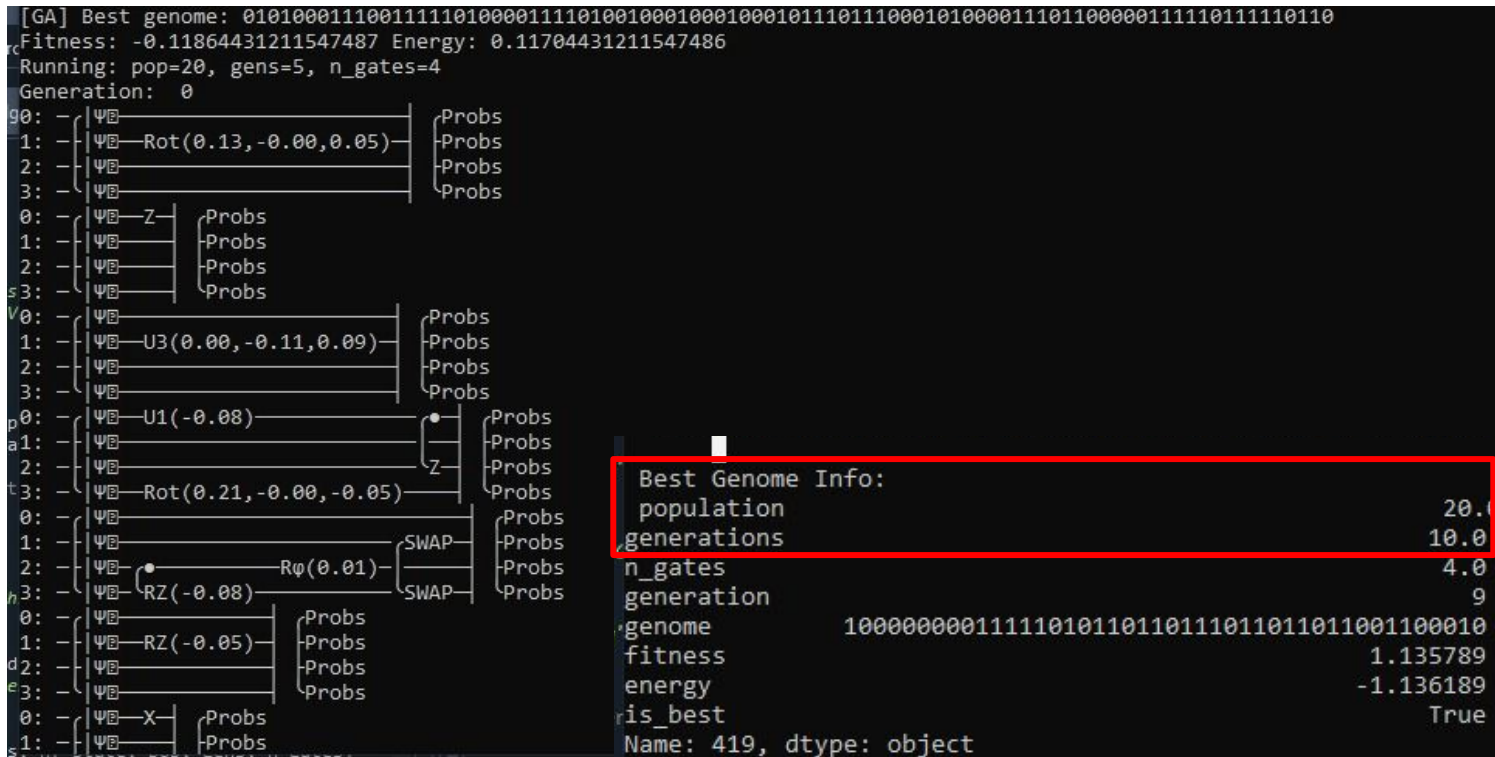


Figure 1: Evolution of the gates and selection of the best gnome

Results

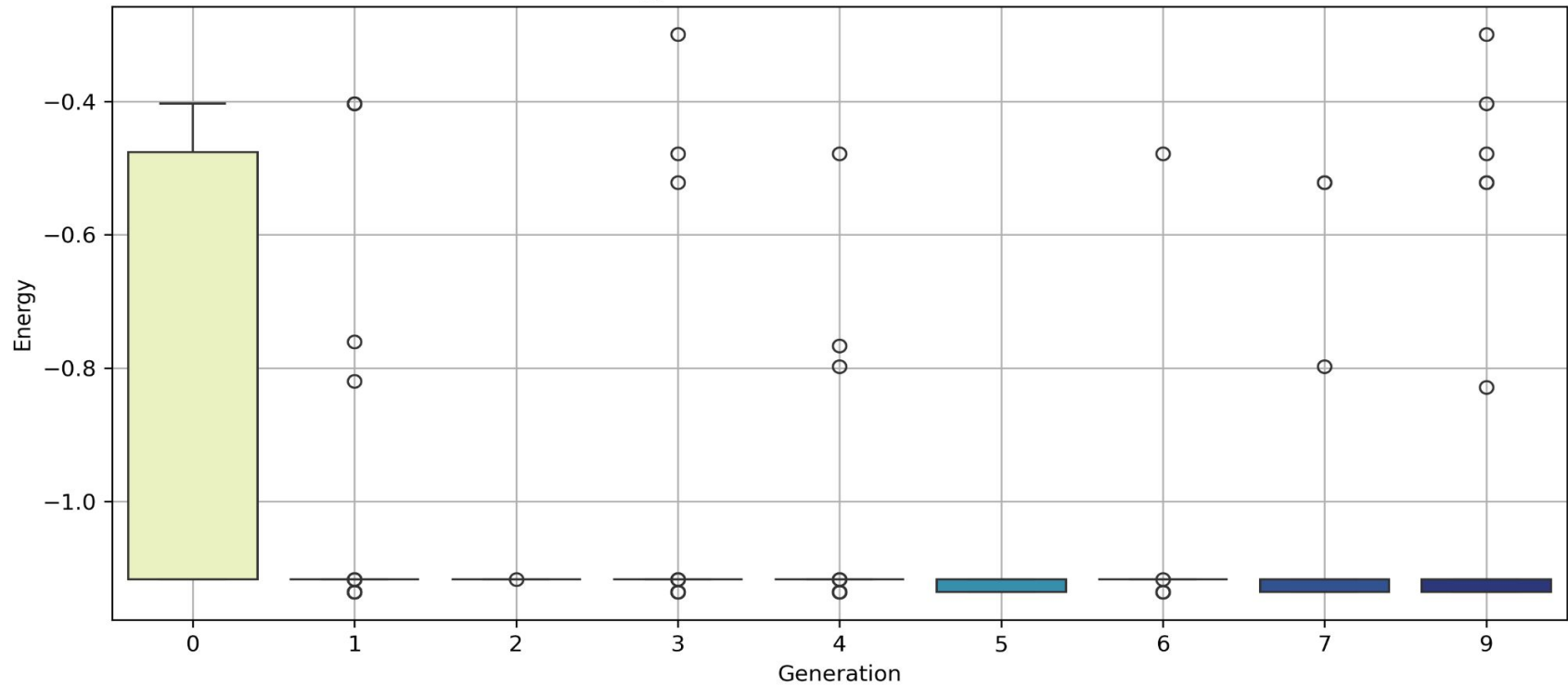


Figure 2: Energy Distribution Across Generations

Results

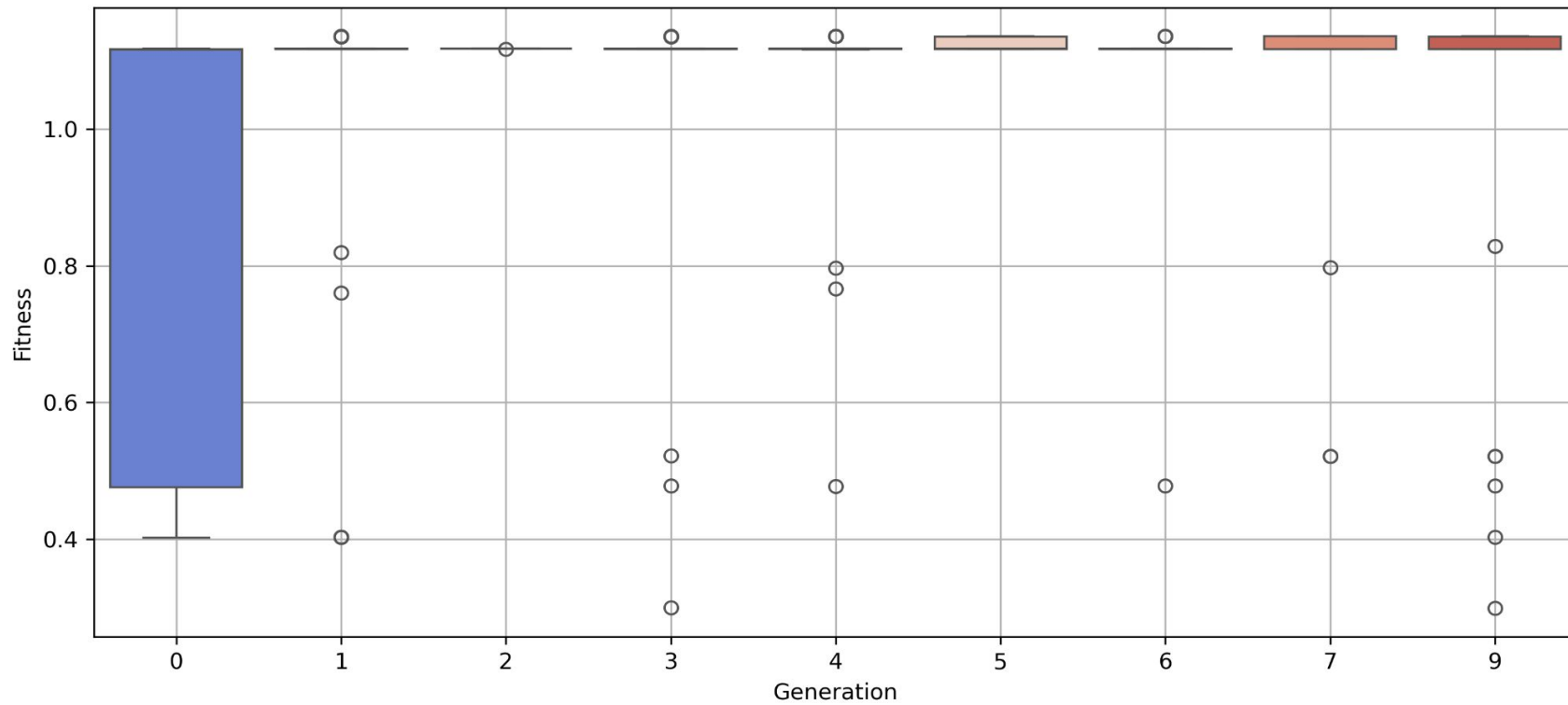


Figure 3: Fitness Distribution Across Generations

Results

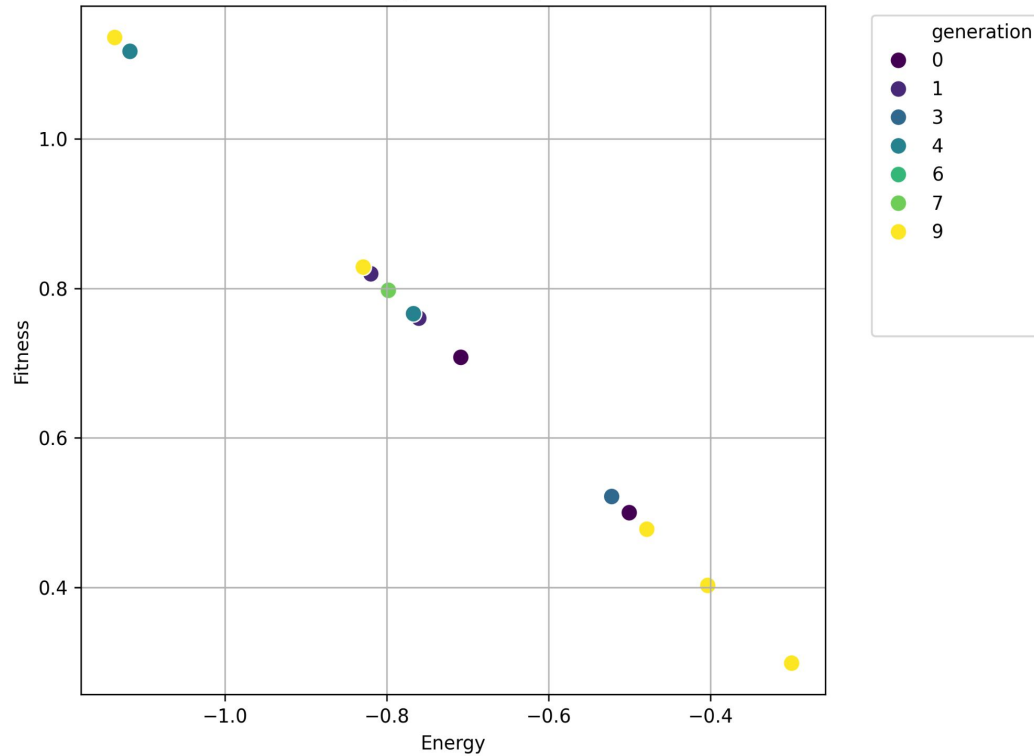


Figure 4: Fitness vs Energy (colored by generation)

Results: GA - qNEAT Benchmarking

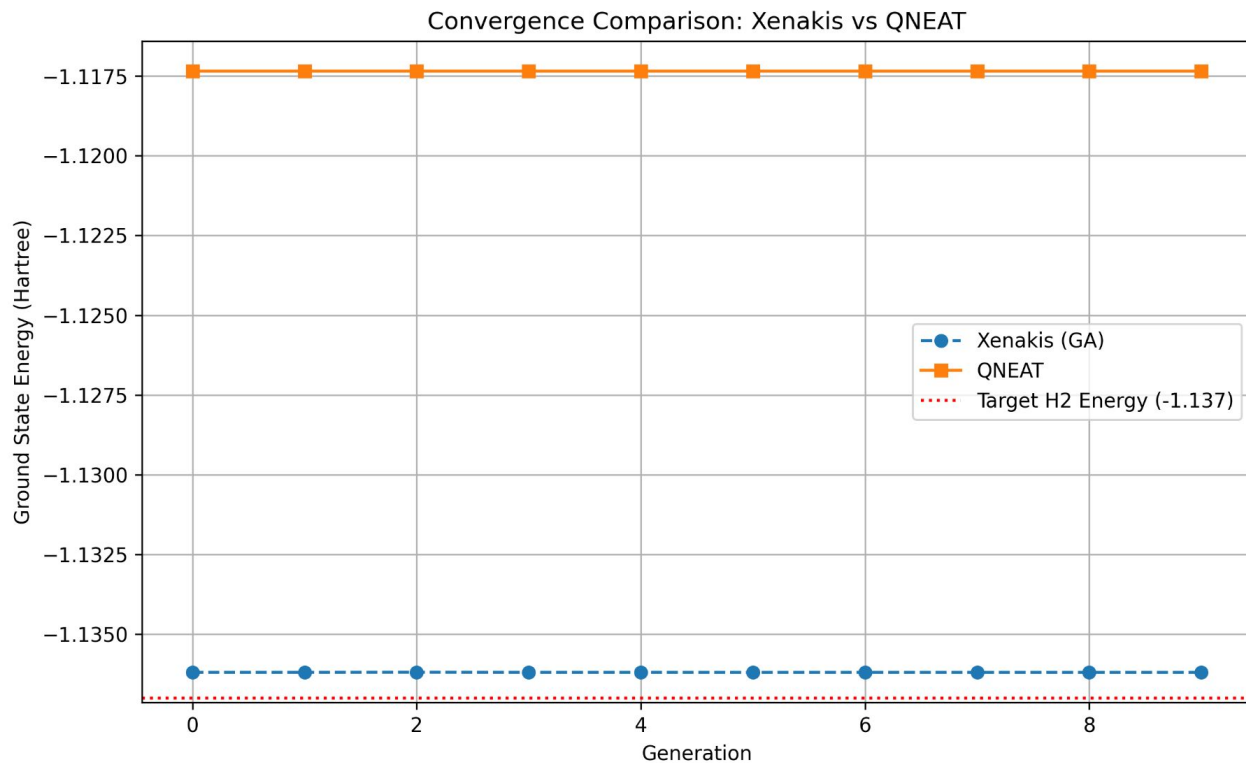


Figure 5: GA vs qNEAT Comparison on Ground State Energy

Results: GA - qNEAT Benchmarking

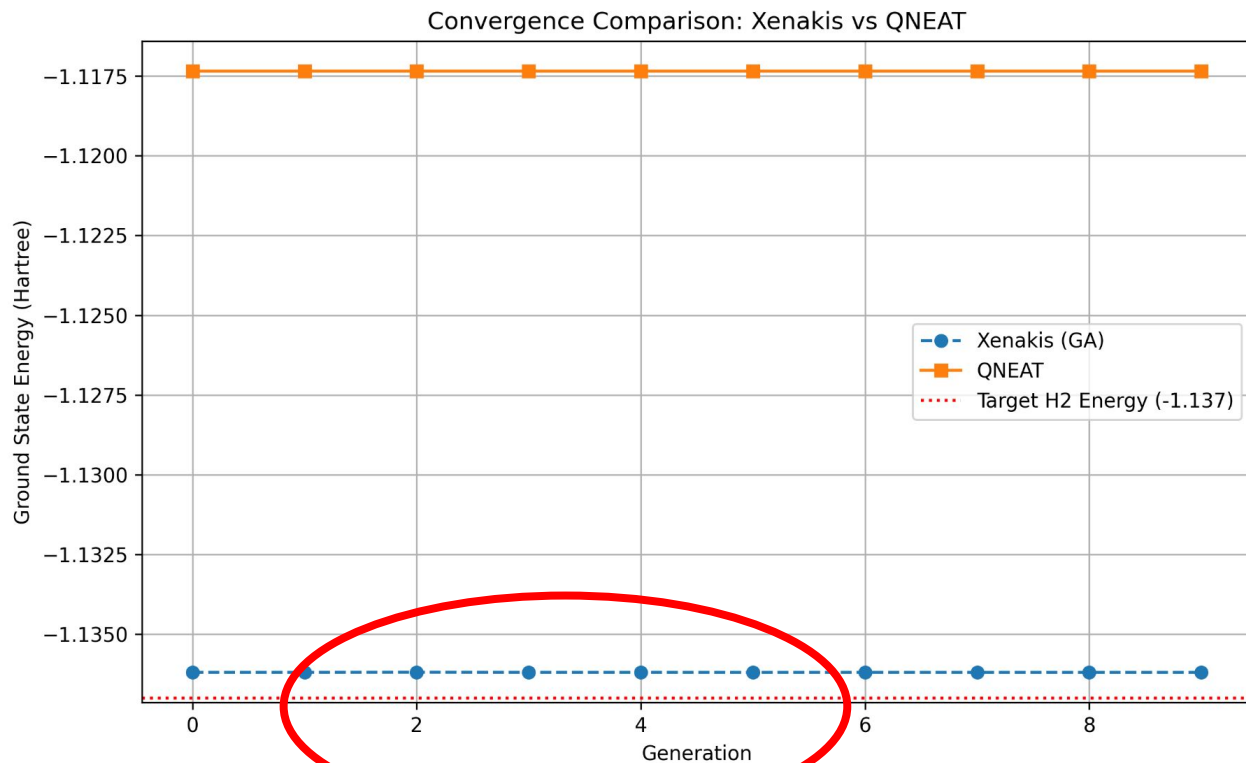


Figure 5: GA vs qNEAT Comparison on Ground State Energy

Results: GA - qNEAT Benchmarking

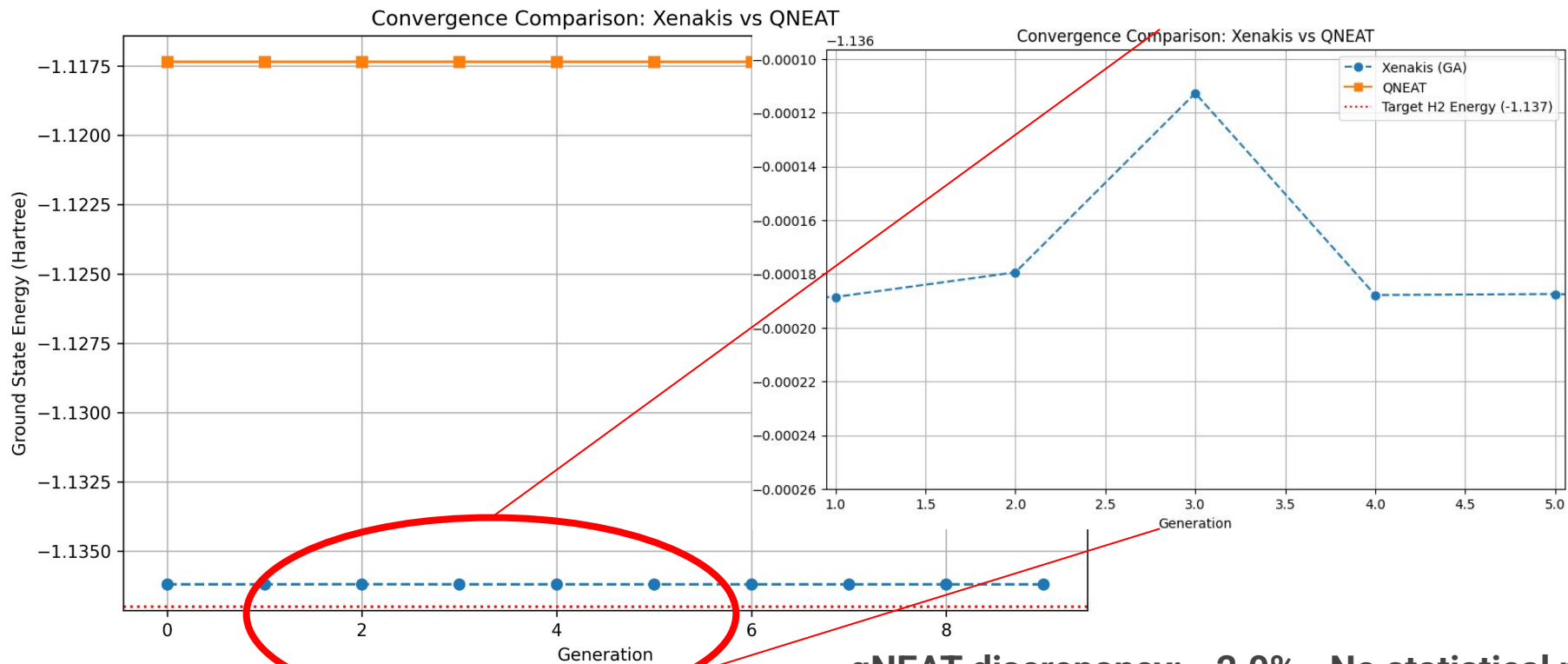


Figure 5: GA vs qNEAT Comparison on Ground State Energy

qNEAT discrepancy: ~2.0% - No statistical noise
GA discrepancy: ~0.1% - Statistical noise

Results: An alternative approach

- Same Genetic Algorithm structure.
- Simpler gates:
Up to 2-qubit gates

Generation 21				
best_fitness	mean_fitness	best_depth	best_2q	unique
1.106684	1.084660	1	0	40

Generation 22				
best_fitness	mean_fitness	best_depth	best_2q	unique
1.106684	1.077308	1	0	40

Generation 23				
best_fitness	mean_fitness	best_depth	best_2q	unique
1.106684	1.072694	1	0	40

Generation 24				
best_fitness	mean_fitness	best_depth	best_2q	unique
1.106684	1.072662	1	0	40

Best genome struct:

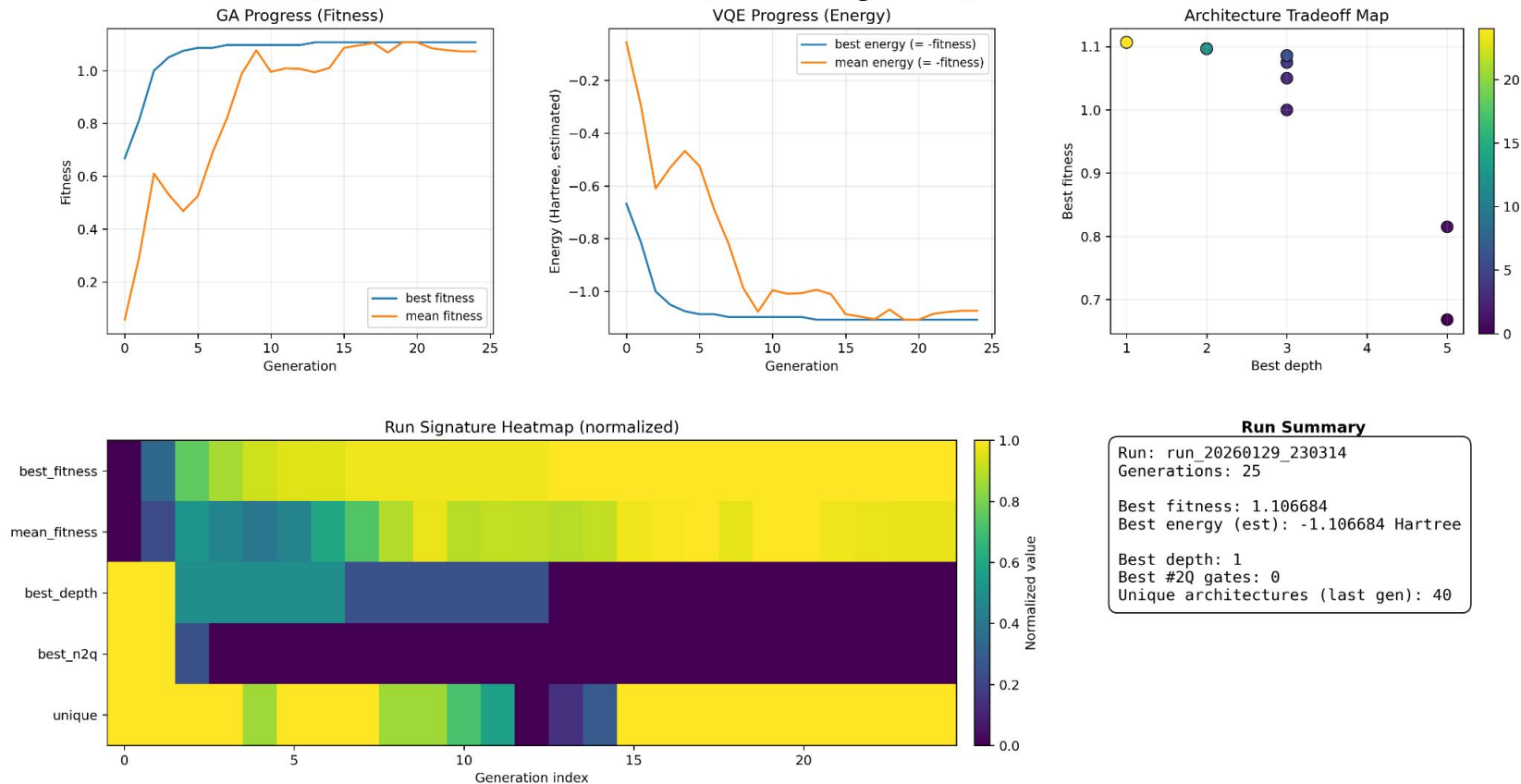
```
{'n_qubits': 4, 'age': 12, 'layers': [[{'name': 'rz', 'wires': [3], 'param': 2.5453324563737576}]]}
```

Last gen stats:

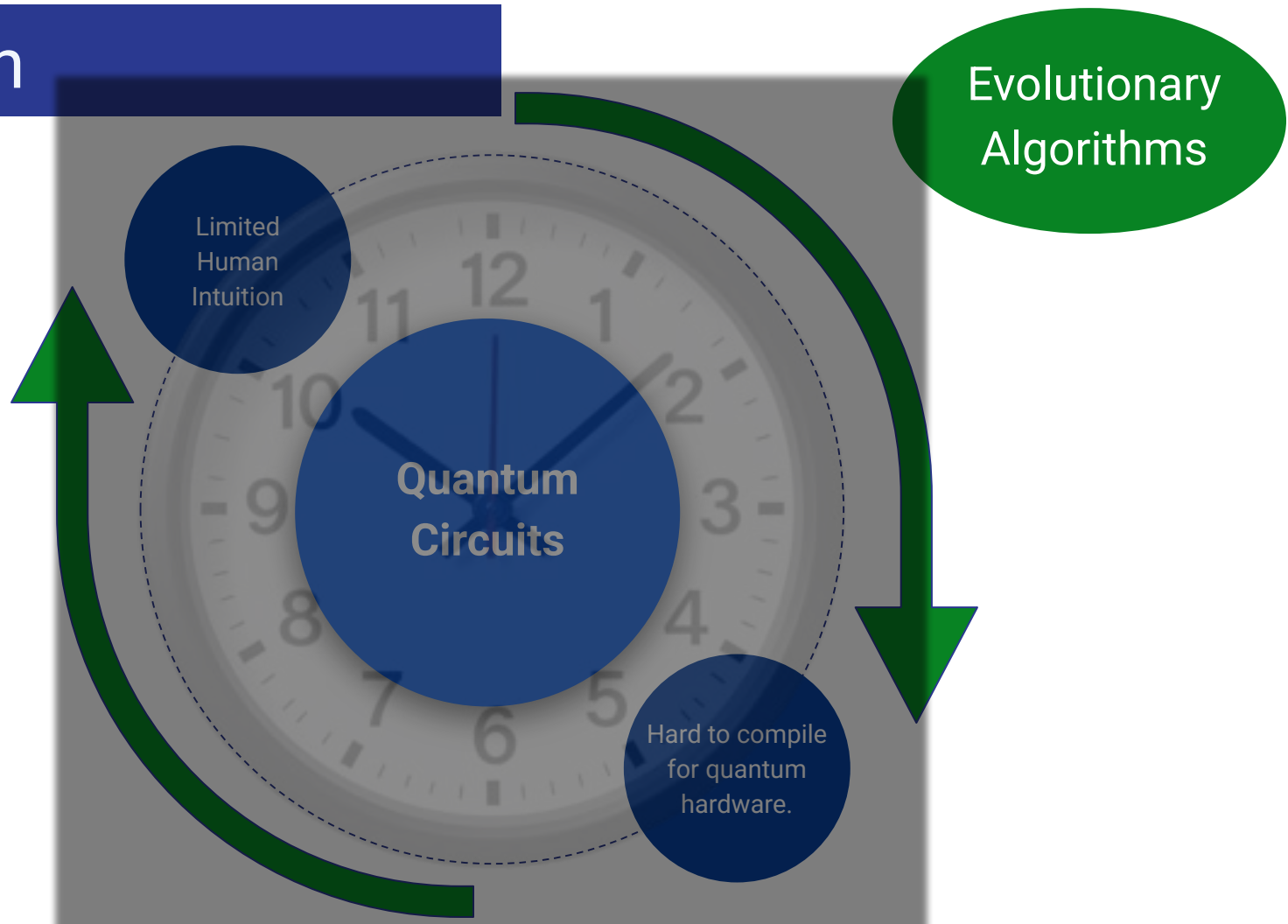
```
{'gen': 24, 'best_fitness': 1.1066843901671173, 'mean_fitness': 1.0726616346424416, 'best_depth': 1, 'best_n2q': 0, 'unique': 40}
```

Results: An alternative approach

Quantum Architecture Search (Genetic Algorithm) — Run Dashboard



Conclusion



Thanks

Solution

Evolutionary
Algorithms

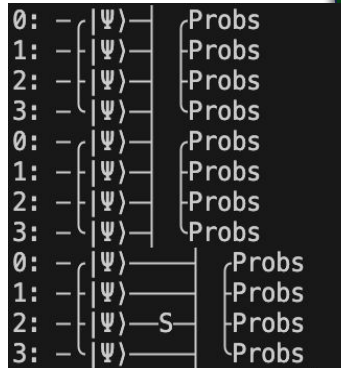
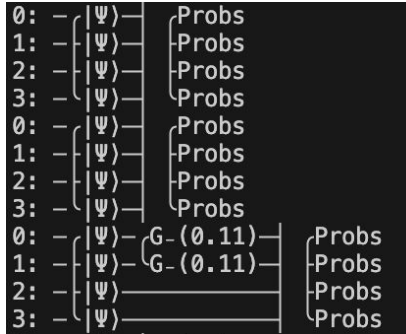
Limited
Human
Intuition

Quantum
Circuits

Hard to
compile for
quantum
hardware.

0:	-	$ \Psi\rangle$		Probs
1:	-	$ \Psi\rangle$		Probs
2:	-	$ \Psi\rangle$		Probs
3:	-	$ \Psi\rangle$		Probs
0:	-	$ \Psi\rangle$		Probs
1:	-	$ \Psi\rangle$		Probs
2:	-	$ \Psi\rangle$		Probs
3:	-	$ \Psi\rangle$		Probs
0:	-	$ \Psi\rangle$		Probs
1:	-	$ \Psi\rangle$		Probs
2:	-	$ \Psi\rangle$	S	Probs
3:	-	$ \Psi\rangle$		Probs

Solution



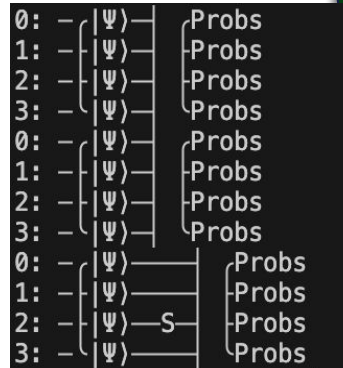
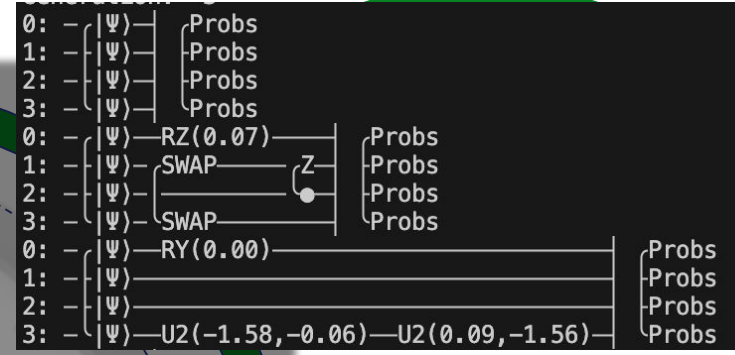
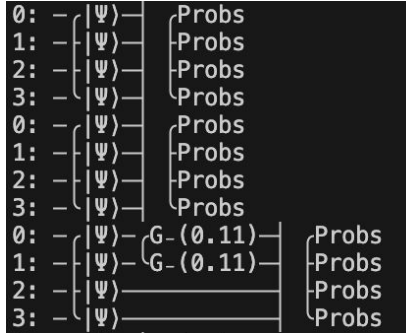
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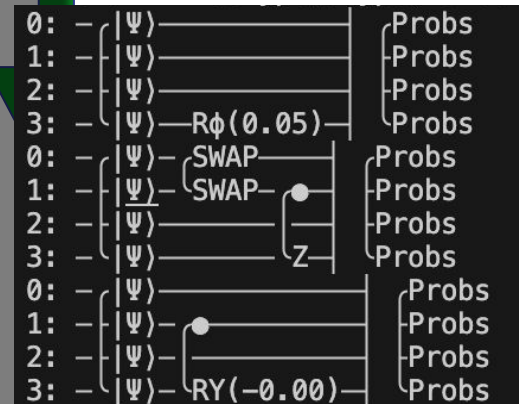
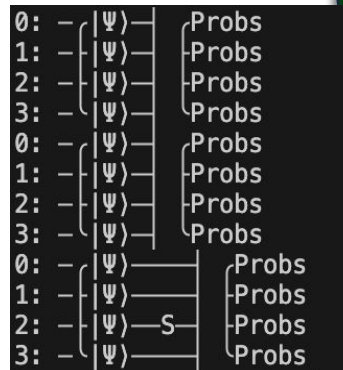
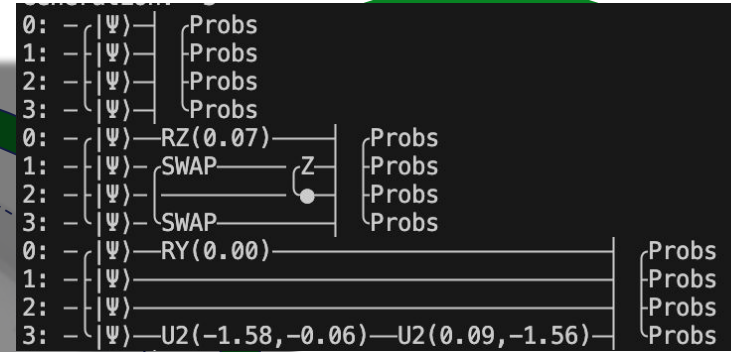
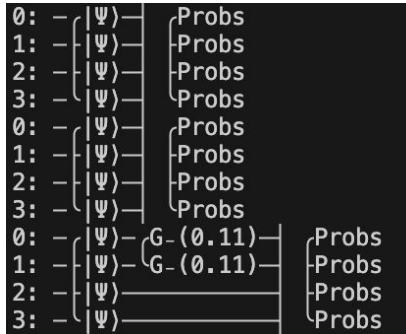


Limited
Human
Intuition

Quantum
Circuits

Hard to
compile for
quantum
hardware.

Solution



Limited
Human
Intuition

Quantum
Circuits

Hard to
compile for
quantum
hardware.

Solution

More premium subscribers

Lorem ipsum dolor sit amet,
consectetur adipiscing elit, sed do
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labore et dolore magna aliqua.

Methodology

Challenge 1

Expand audien

Challenge 2

Up 30-day actives

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veniam, quis nostrud
exercitation

- Duis aute irure dolor
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Challenge 3

Increase conversion

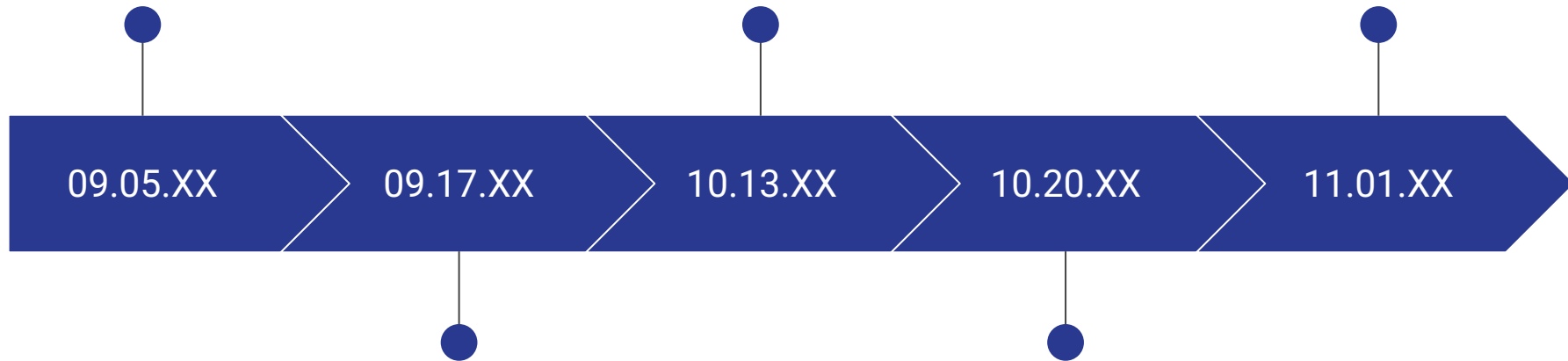
Excepteur sint occaecat
cupidatat non proident,
sunt in culpa qui officia
deserunt mollit anim id
est laborum.

Implementation

Lorem ipsum dolor sit
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Lorem ipsum dolor sit
amet, consectetur
adipiscing elit

Lorem ipsum dolor sit
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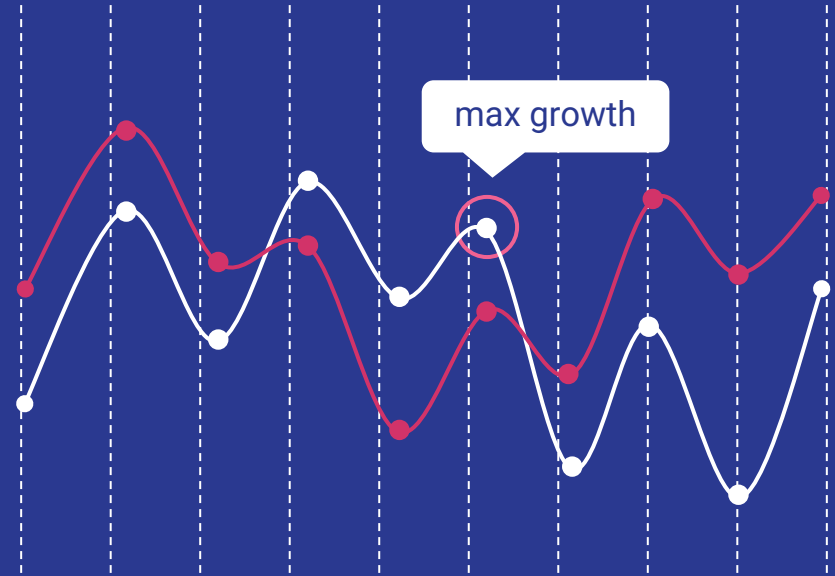


Lorem ipsum dolor sit
amet, consectetur
adipiscing elit

Lorem ipsum dolor sit
amet, consectetur
adipiscing elit

Impact

XX% sales increase



[GA] Best genome: 0101000111100111110100001111010010001000101110111000101000011101100000111110111110110
 Fitness: -0.11864431211547487 Energy: 0.11704431211547486
 Running: pop=20, gens=5, n_gates=4
 Generation: 0

g0: — Ψ ———— | Probs
 1: — Ψ — Rot(0.13, -0.00, 0.05) — | Probs
 2: — Ψ ———— | Probs
 3: — Ψ ———— | Probs

0: — Ψ — Z — | Probs
 1: — Ψ ———— | Probs
 2: — Ψ ———— | Probs
 3: — Ψ ———— | Probs

V0: — Ψ ———— | Probs
 1: — Ψ — U3(0.00, -0.11, 0.09) — | Probs
 2: — Ψ ———— | Probs
 3: — Ψ ———— | Probs

p0: — Ψ — U1(-0.08) ———— | Probs
 a1: — Ψ ———— | Probs
 2: — Ψ ———— | Probs
 t3: — Ψ — Rot(0.21, -0.00, -0.05) — | Probs

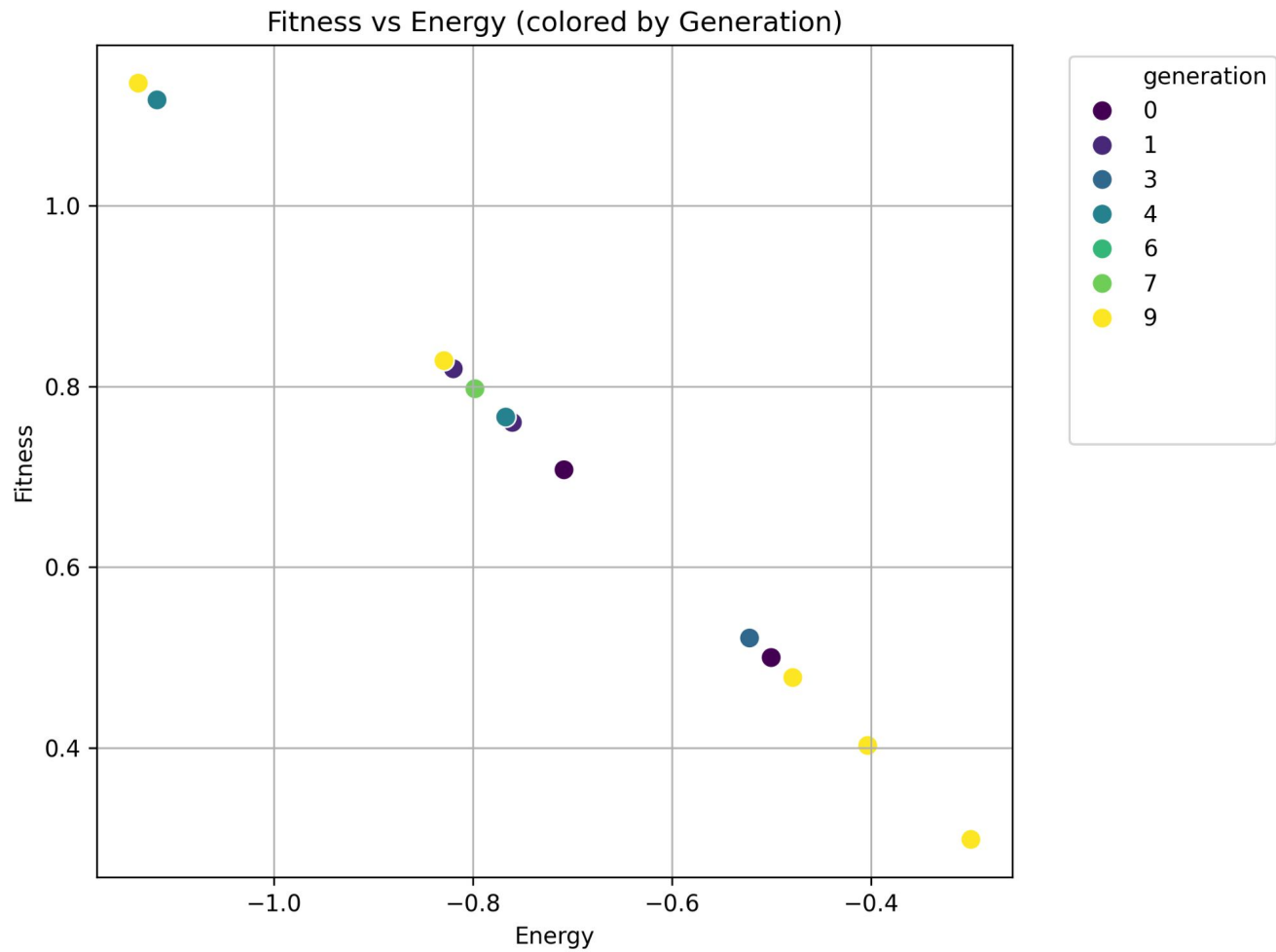
0: — Ψ ———— | Probs
 1: — Ψ ———— | Probs
 2: — Ψ — $R\phi(0.01)$ — SWAP — | Probs
 h3: — Ψ — (RZ(-0.08)) — SWAP — | Probs

0: — Ψ ———— | Probs
 1: — Ψ — RZ(-0.05) — | Probs
 d2: — Ψ ———— | Probs
 e3: — Ψ ———— | Probs

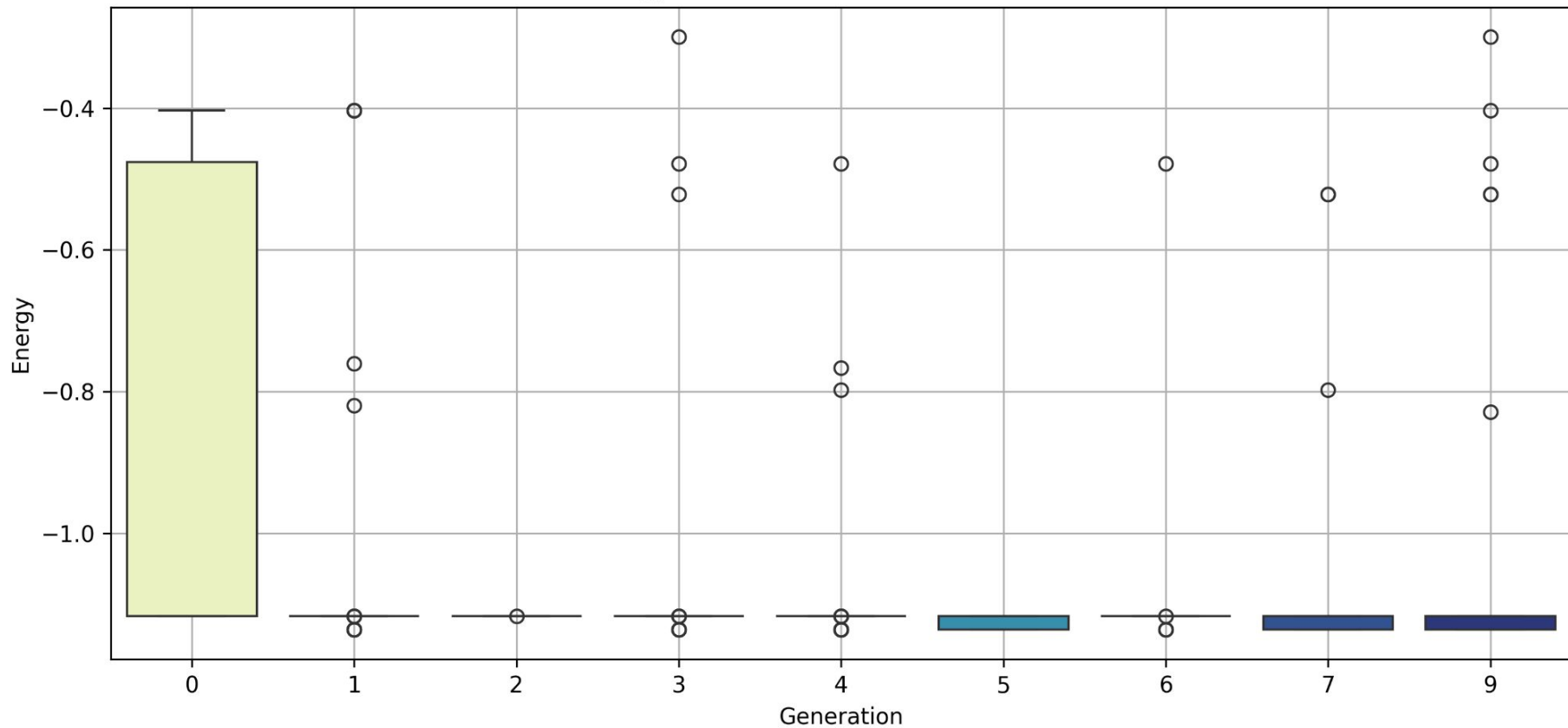
0: — Ψ — X — | Probs
 1: — Ψ ———— | Probs

Best Genome Info:

population	20.0
generations	10.0
n_gates	4.0
generation	9
genome	10000000011111010110110111011011011001100010
fitness	1.135789
energy	-1.136189
is_best	True
Name: 419, dtype: object	



Energy Distribution Across Generations



Fitness Distribution Across Generations

